Kian (Mohammad) Kianpisheh

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Summary: Last year PhD student at the University of Toronto specializing in Applied AI and Machine Learning, with 6+ years of research and hands-on experience. Experienced in designing, developing, and deploying intelligent systems across human activity recognition, computer vision, and audio analysis domains. Skilled in building end-to-end machine learning solutions, leveraging multimodal data, self-supervised learning, and LLMs.

EXPERIENCE

University of Toronto 2018-2025

Research Assistant — Applied AI

- Fine-tuned lightweight LLMs (Phi-3.5 Mini, and TinyLlama) using HuggingFace Transformers to power a context-aware human activity recognition agent.
- Leveraged user feedback as a Retrieval-Augmented Generation (RAG) resource to personalize human activity recognition and improve predictions with Hugging Face Transformers.
- Designed personalized intelligent systems using multimodal machine learning techniques, including self-supervised and one-shot learning, for adaptive behavior modeling.
- Worked with motion, vision, audio, and other sensor data to model human behavior and context.
- Focused on adaptive and user-aware AI/IoT applications, combining signal processing with deep learning approaches.

Mathsronauts 2023-2025

Machine Learning Engineer (PyTorch, OpenCV)

- Developed a platform that allowed students to interactively learn and experiment with computer vision and machine learning concepts.
- Built the user interfaces for the teaching platform for students using React.js, TypeScript, and Redux.
- Implemented backend APIs using Node.js and Express.js to support user authentication and dynamic content delivery.

Paya Vision 2013-2015

Machine Learning Engineer (TensorFlow, OpenCV, OpenCL, Scikit-Learn)

- Developed a content-based video retrieval system based on Topic Models for traffic surveillance videos.
- Developed an anomaly detection system for traffic surveillance videos using Topic Models.
- Designed and implemented a license plate recognition system.

EDUCATION

University of Toronto Sep 2018-Present

Ph.D. candidate in Computer Science, DGP lab

Personalized Ubiquitous Intelligent Systems

Advisor: Prof. Khai Truong

Sharif University of Technology 2015-2017

M.Sc. in Electrical Engineering, (Computer Vision, Scene Understanding)
Advisor: Prof. Mohammad Sharifkhani

Shahed University 2008-2013
B.Sc. in Electrical Engineering, (Image Processing)
Advisor: Prof. Alireza Behrad

RESEARCH AND PUBLICATIONS

Kianpisheh M, Mariakakis A, Truong KN. SAHARA: Self-supervised Approach for Human Activity Recognition based on Everyday Audio Events. (under review)

Kianpisheh M, Mariakakis A, Truong KN. exHAR: An Interface for Helping Non-Experts Develop and Debug Knowledge-based Human Activity Recognition Systems. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2024 Mar 6;8(1):1-30. (link)

Kianpisheh M, Li FM, Truong KN. Face recognition assistant for people with visual impairments. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2019 Sep 9;3(3):1-24. (link)

Kianpisheh M, Content-based Video Retrieval in Traffic Videos using Latent Dirichlet Allocation Topic Model. 2017 Sep, arxiv. (link)

SELECT LIST OF TECHNICAL PROJECTS

SAHARA: An audio-based self-supervised human activity recognition system. 2025
The system automatically clusters repetitive audio events and characterizes the user's activity based on those detected events.

IMUdio: A smartwatch human activity recognition system based on audio and IMU sensor data. 2024
A lightweight deep learning model that fuses audio and IMU data embeddings to recognize human activities, with inference performed directly on a smartwatch.

exHAR: An Interface for Helping Non-Experts Develop and Debug Human Activity Recognition Systems. 2024
An explainable framework that enables users to personalize sensor-based human activity recognition systems.

SensorLab: A data visualization tool for wearable devices and smartphone IMU sensor data. 2023
A visualization web application that shows the spectrogram of IMU (i.e., accelerometer and gyroscope) sensor data.

ActivityLab: A data visualization tool for wearable devices and smartphone sensors data. 2022
A web-based visualization application that depicts human activity datasets as a chronological sequence of events. Users can program the system in If-This-Then-That format.

Face recognition assistant for people with visual impairments. 2020

A smart wearable camera that automatically captures the faces of people the user interacts with. It filters out low-quality images (poor lighting, extreme head poses, motion blur) to ensure reliable face classification.

CBVR: Content-based Video Retrieval using Latent Dirichlet Allocation Topic Model. 2017
The system detects traffic patterns using Topic Models. A GUI allows users to search for specific patterns efficiently.

TEACHING EXPERIENCE

University of Toronto

- Foundation of Computer Vision
- Programming on the Web (Lead TA) (4x)
- Introduction to Computer Science (>4x)
- Computational Thinking
- Image Understanding (2x)
- Data Structures and Analysis
- Introduction to Programming (>4x)
- Great Ideas in Computing

SKILLS

Machine Learning: PyTorch TensorFlow Hugging Face OpenCV
Web development: React React Node JavaScript TypeScript
Design: Figma Illustrator
Mobile Development: Android (Java)

REFERENCES

Khai N. Truong (Professor, Department of Computer Science, University of Toronto)

Alex Mariakakis (Asistant Professor, Department of Computer Science, University of Toronto)